

# Reemika Subrata Das

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## PROFESSIONAL SUMMARY

Data Analytics and Applied AI professional with 5+ years of combined experience in MIS reporting, business intelligence, stakeholder requirements gathering, predictive modeling, and AI application development. Recently earned an M.S. in Business Analytics with an AI/ML concentration from Santa Clara University. Hands-on experience building ETL/ELT pipelines, Text-to-SQL and RAG applications, forecasting models, and machine-learning pipelines using Advanced Excel, Python, SQL, Tableau, PySpark, LangChain, and AWS EC2.

## TECHNICAL SKILLS

**Data Analysis & Programming:** Python, SQL, Advanced Excel, Pandas, NumPy, data extraction, data cleaning, data validation, exploratory data analysis, ad hoc analysis, data reporting, stakeholder requirements gathering, process documentation

**BI/Data Engineering:** ETL/ELT pipelines, Databricks, PySpark, Delta Lake, Unity Catalog, Auto Loader, MySQL, schema validation, Tableau, Matplotlib, dashboards, KPI reporting

**Applied AI & LLMs:** Generative AI, Retrieval-Augmented Generation (RAG), Text-to-SQL, prompt engineering, LLMops, LangChain, FAISS, Hugging Face, LangSmith

**Machine Learning:** Model development, feature engineering, supervised learning, time-series forecasting, model evaluation, Scikit-learn, Statsmodels, TensorFlow, Keras

**Cloud & AI-assisted Tools:** AWS EC2, Git, GitHub, Codex, Claude Code, REST APIs, Streamlit, FastAPI

## PROFESSIONAL EXPERIENCE

### Santa Clara University

Research Assistant | Dec 2025 - Present

- Developed, tested, and deployed a scalable Python-based AI-powered provenance assistant on AWS EC2, integrating Large Language Models (LLMs) with Flask and FastAPI to generate inspectable MySQL queries across 41K+ artwork and 70K+ image records
- Applied prompt engineering, context engineering, and bounded conversation-memory management while keeping query execution and pagination outside the LLM context to securely process 111K+ records without exposing raw database rows to the model
- Engineered a secure, and testable LLM application architecture using SQLGlot validation, approved-table authorization, SELECT-only enforcement, semantic matching, query controls, and multi-model fallback, improving reliability across 111K+ structured records
- Building an AI evaluation and observability pipeline using Python, Pandas, SQL, and Tableau-ready data models to monitor LLM token consumption, latency, provider performance, and cost per request for model quality, performance, and cost optimization

### Adobe (with Santa Clara University)

Data Scientist Practicum | Jan 2025 - Jun 2025

- Applied statistical analysis and machine learning techniques to develop, evaluate, and benchmark ARIMA, feed-forward neural network, RNN, and LSTM models, achieving 2.4% MAPE for subscription-seat demand forecasting
- Built a reusable Python data-processing and ML pipeline across 6,400+ structured daily records, engineering lag, rolling-window, fiscal-calendar, holiday, growth, and Fourier features to model trend and seasonality
- Prepared and validated modeling datasets using Python, Pandas, NumPy, statistics, and probability, applying missing-value treatment and ADF, and KPSS tests to assess stationarity and support reliable model development

### XL Dynamics India Pvt Ltd

Process Executive MIS | Nov 2021 - Aug 2024

- Prepared 30+ MIS reports for senior management, finance, and marketing teams by extracting, cleaning, validating, and transforming structured mortgage data using Excel VLOOKUP, PivotTables, logical formulas, and filters
- Created impactful Excel dashboards and interpreted data for reporting across 7-8 cross-functional teams to deliver decision-ready analysis
- Performed ad hoc analysis to translate business requirements into analytical solutions and reduced knowledge gaps and escalation by creating a Standard Operating Procedure (SOP) to carry out routine operations

### MP Financial Advisory Services LLP

Credit Risk Analyst | Jun 2021 - Oct 2021

- Supported lending and advisory decisions, measured by 5 client credit evaluations, by building Excel-based financial models and summarizing creditworthiness, risk exposure, and recommendation drivers

### Trio Logistics Pvt Ltd

Account Assistant | Jun 2018 - Jun 2019

- Improved tax-filing readiness for 50+ clients by preparing reconciliation reports, monthly summaries, and accurate financial records

## SELECTED PROJECTS

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### Uber vs Lyft Fare Analytics Platform | Databricks ELT, Tableau BI & Machine Learning

- Engineered a governed Databricks medallion ELT pipeline for 699,347 fare and weather records using PySpark, Auto Loader, Delta Lake, and Unity Catalog; validated and retained 92% of fare quotes and achieved 100% endpoint-weather matching
- Analyzed 425,321 comparable fare quotes across 72 directional routes through an interactive Tableau dashboard, identifying lower median price per mile for Lyft in Standard and XL tiers and for Uber in both Premium Black tiers
- Developed leakage-safe regression models in PySpark ML; selected Gradient-Boosted Trees over the tuned Random Forest for greater efficiency, improving MAE by 32% over Linear Regression and achieving a \$1.19 test MAE and 0.9625 R<sup>2</sup>

### SQL Challenge Journal & Notebook Publisher | MySQL, PostgreSQL, GitHub API & GitHub Actions | [Live Application](#)

- Built and launched a public SQL Notebook Publisher that imports HackerRank and DataLemur challenge details, generates standardized Markdown documentation, and publishes solutions to user-selected GitHub repositories
- Solved and documented 70+ SQL challenges using joins, CTEs, subqueries, aggregations, ranking, and window functions across business, product, finance, and user-behavior scenarios
- Designed an automated publishing workflow covering source URL validation, Markdown preview, and secure GitHub API commits, reducing manual publishing effort by approximately 80% and automating repository metrics through GitHub Actions

### HealthMate | Conversational RAG Fitness & Nutrition Guide

- Designed and developed a full-stack Retrieval-Augmented Generation (RAG) solution over eight curated knowledge documents using React, TypeScript, Python, FastAPI, LangChain, Hugging Face embeddings, FAISS vector search, semantic retrieval, and Groq LLM inference to produce grounded, cited responses
- Created an AI evaluation methodology with 36 benchmark test cases and LangSmith/LLM-as-judge workflows to measure correctness, relevance, groundedness, retrieval quality, and end-to-end RAG performance

## EDUCATION

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Santa Clara University - M.S. in Business Analytics, AI/ML Concentration | GPA: 3.71/4.00 | Sept 2024 - Dec 2025

ITM Business School - Post Graduate Diploma in Management | GPA: 3.02/4.00 | Jun 2019 - Jun 2021

University of Mumbai - Bachelor's in Management Studies | GPA: 3.70/4.00 | Jun 2015 - Apr 2018